Bachar Bouguerra Zina

AI/ML Engineer — LLM & RAG systems, predictive modeling, applied deep learning

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Profile

Applied AI engineer (INSAT, 2026) who takes AI projects from data to deployed service. Experience across Germany, France, Canada and Tunisia building RAG assistants, fine-tuned LLMs, time-series and physics-informed models, and data pipelines. Comfortable owning a project end to end: scoping with stakeholders, building, evaluating and shipping with FastAPI and Docker.

Experience

Machine Learning Researcher — Université Laval, Québec City, Canada · 07/2026 – 10/2026

- Built predictive ML models estimating physical, chemical and biological soil indicators from heterogeneous spectroscopic data.
- Harmonised Vis-NIR, MIR, XRF and LIBS sources into a validated database; evaluated model robustness against laboratory reference measurements.
- Applied statistical quality control to detect non-conforming measurement series.

ML/AI Engineer (end-of-studies internship) — Unisphere GmbH, Konstanz, Germany · 02/2026 – 06/2026

- Built and deployed a production RAG assistant (Gemini, Qdrant) with semantic search, intent filtering and conversation state.
- Built a data pipeline over 17,000 UAV flights: time-series structuring, feature engineering, reliability scoring, monitoring dashboards.
- Trained a Temporal Convolutional Network with Viterbi decoding to classify flight phases under physically valid transitions.
- Derived a physics-informed power and true-airspeed model, cutting prediction error by 57% versus a data-driven baseline.

- Deployed FastAPI and Docker services with semantic caching to balance latency and inference cost.

ML/AI Engineer Intern — YONNOV'IA, Marseille, France (remote) · 06/2025 – 08/2025

- Built a RAG assistant with LangChain and Ollama integrated into Odoo 17 for employee questions (company info, meetings, holidays).
- Fine-tuned Phi-3 Mini with LoRA and 4-bit quantisation: over 90% accuracy, responses under 5 seconds.
- Evaluated factual consistency, relevance and hallucination rate while iterating on retrieval and prompts.

Data & NLP Intern — ACOBA, Monastir, Tunisia · 07/2024 – 09/2024

- Scraped real-estate listings from Leboncoin with Python and API integrations into a structured dataset.
- Cleaned and structured the data with NLP techniques for the chatbot's knowledge base.
- Built a Rasa chatbot from scratch (intents, entities, dialogue flows) answering filtered property queries, with about 80% query-conversion success.

Logistics Optimisation Intern — Dräxlmaier, Monastir, Tunisia · 08/2023 – 09/2023

- Applied machine learning to logistics flow optimisation, contributing to about a 20% gain in operational efficiency.

Robotics Instructor — EIF Menzah 1, Tunis · 10/2024 – 03/2025 — taught robotics and programming to children aged 8–12.

Selected projects

- **Car-Finder** — LLM engine converting English/French questions into schema-compliant SQL with rule-based validation; 97% precision.
- **Fracture Detection App** — Flutter app with a CNN (TensorFlow Lite, on-device) detecting bone fractures in X-rays; 93% accuracy.
- **NASA Turbofan RUL Predictor** — LSTM remaining-useful-life model on NASA C-MAPSS sensor time series, served with Flask.
- **Virtual Interviewer** — upload a CV and job description to get tailored interview questions and feedback.

- **Tunisian History RAG** — retrieval-augmented QA over documents with embeddings and FAISS vector search.

Education

- **Engineering Degree in Applied Artificial Intelligence** — INSAT, Tunis · 2021 – 2026 · thesis: *AI-Enhanced UAV Performance Modeling* (Unisphere GmbH)
- **Mathematics Baccalaureate, highest honours (17.97/20)** — Pioneer High School of Monastir · 2021

Awards

- **1st place** — IEEE Region 8 Industrial Bootcamp 4.0, IEEE IAS INSAT (02/2025): AIgroFarm, AI-monitored sustainable farming with computer vision for fruit maturity.
- **2nd place** — Orbit 1.0 Ideathon, IEEE CS INSAT (05/2025): EcoAI-Train, tracking and reducing energy waste in model training.
- **1st place** — NRTF Hackathon, IEEE RAS INSAT (04/2024): Dhawwini, solar and AI-controlled turbine energy for non-electrified areas.

Certificates

Machine Learning Specialization (Coursera) · Computer Vision for Industrial Inspection (NVIDIA) · Agile Scrum Methodology (The Way Center)

Skills

- **LLM & GenAI:** RAG, LangChain, Ollama, Gemini API, Qdrant, FAISS, LoRA / 4-bit fine-tuning, Hugging Face Transformers, Rasa
- **Machine learning:** PyTorch, TensorFlow/Keras, scikit-learn, time-series (TCN, LSTM), CNNs / computer vision, physics-informed modeling, spectroscopic data
- **Engineering:** Python, SQL, Java, FastAPI, Flask, Docker, PostgreSQL, Git, Linux, web scraping, n8n

Languages

Arabic (native) · French (fluent) · English (fluent) · German (basic) · Spanish (basic)